

Book 2: Chapter 4 – Optimal Clustering (ONC)

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Table of Contents

- 1 Overview & Motivation
- 2 The ONC Algorithm
- 3 Financial Applications
- 4 Summary & Conclusion

Chapter 4 Overview: Unsupervised Learning in Finance

Applications of Clustering in Quant Finance

- **Asset Taxonomy:** Grouping assets by empirical return codependency rather than discretionary sector labels.
- **Feature Reduction:** Grouping collinear features to eliminate substitution effects in feature importance.
- **Portfolio Construction:** Structuring covariance matrices for **Nested Clustered Optimization (NCO)**.

The Core Challenge

- Financial distance matrices derived from correlation or Variation of Information contain significant sampling noise.
- Standard clustering algorithms (like k -means) require specifying k ex-ante and suffer under multicollinearity and noise.
- **ONC Solution:** An automated, noise-robust clustering algorithm based on Random Matrix Theory and Silhouette score quality metrics.

Why Standard k -Means Fails in Quantitative Finance

Three Fundamental Flaws of Standard k -Means

- 1 **Subjective Heuristics (The Elbow Method):** The "elbow" location on an inertia plot is visually ambiguous and non-deterministic under financial noise.
- 2 **Initialization Instability:** Random centroid initialization yields different cluster partitions across random seeds, leading to high allocation turnover.
- 3 **Silhouette Maximization under Multicollinearity:** Maximizing average Silhouette score alone overfits to noisy sub-clusters and fails to identify the true underlying block-diagonal structure.

The Need for ONC

Optimal Number of Clusters (ONC) replaces subjective heuristics with a rigorous, two-stage optimization algorithm that evaluates cluster partition quality via t -statistics.

Topic 4.1: The ONC Base Stage & Partition Quality

Silhouette Score S_i

For observation i , let a_i be the mean intra-cluster distance and b_i be the mean nearest-cluster distance:

$$S_i = \frac{b_i - a_i}{\max(a_i, b_i)} \in [-1, 1]$$

Partition Quality Metric q

ONC evaluates a clustering scheme by the t -statistic of its Silhouette scores:

$$q = \frac{E[\{S_i\}]}{\sqrt{V[\{S_i\}]}}$$

Higher q indicates tight, well-separated clusters with low variance.

Base Stage Procedure

- For each candidate $k \in [1, N - 1]$:
- Run k -means on the distance matrix D .
- Compute q_k . Select $k^* = \arg \max_k q_k$.
- Yields initial partition $C^* = \{C_1, \dots, C_{k^*}\}$.

Topic 4.2: The ONC Recursive Stage

Identifying Low-Quality Clusters

The base stage may group distinct sub-clusters together. ONC evaluates each cluster $C_k \in C^*$ individually:

$$q_k = \frac{E[\{S_i\}_{i \in C_k}]}{\sqrt{V[\{S_i\}_{i \in C_k}]}}$$

If $q_k < \bar{q}$ (mean quality across all clusters), cluster C_k is flagged for recursive re-clustering.

Recursive Splitting

- Apply ONC base stage exclusively to items in C_k .
- If sub-clustering yields higher quality $q'_{C_k} > q_k$, accept the split into sub-clusters.

Global Re-allocation

- After all recursive splits, re-assign each item to its nearest cluster centroid.
- Consolidate redundant clusters if centroid distances drop below threshold.

Topic 4.3: Step-by-Step ONC Algorithm Summary

The Complete 5-Step ONC Workflow

- 1 **Distance Matrix Creation:** Compute distance matrix D from correlation (d_ρ) or Variation of Information ($\widetilde{VI}$).
- 2 **Base Clustering Search:** Test $k \in [2, N - 1]$, run k -means, compute q_k , select $k^* = \arg \max q_k$.
- 3 **Quality Filtering:** Calculate mean quality $\bar{q}$. Identify sub-average clusters where $q_k < \bar{q}$.
- 4 **Recursive Subdivision:** Recursively apply ONC to low-quality clusters and accept splits that improve local q .
- 5 **Centroid Consolidation:** Re-assign all elements to closest cluster centroids and consolidate overlapping clusters.

Topic 4.4: Application 1 – Clustered Feature Importance

The Substitution Effect Problem

When predictive features are highly correlated (e.g., multiple momentum indicators), standard tree-based MDI and MDA split importance among redundant variables, diluting true signal and inflating noise variables.

Clustered MDI / MDA Solution

- Run ONC on feature distance matrix $D(X_i, X_j)$ to group collinear features into K clusters.
- Compute feature importance at the **cluster level** rather than individual feature level.

Key Advantage

Neutralizes multicollinearity substitution without losing feature interpretability (unlike PCA basis transformations).

Topic 4.5: Application 2 – Nested Clustered Optimization (NCO)

Taming Markowitz's Curse with ONC

Covariance matrix inversion V^{-1} explodes when assets are highly correlated within sectors. NCO uses ONC to isolate condition numbers inside sub-matrices.

The 3-Step NCO Architecture

- ① **Cluster Partitioning:** Use ONC on asset correlation matrix C to form K asset clusters.
- ② **Intracuster Allocation:** Calculate optimal weight vector ω_k^{intra} within each cluster k independently:

$$\omega_k^{\text{intra}} = \frac{V_k^{-1} a_k}{a_k' V_k^{-1} a_k}$$

- ③ **Intercluster Allocation:** Reduce full covariance matrix to intercluster covariance $V_{\text{inter}} = W_{\text{intra}}' V W_{\text{intra}}$, then compute intercluster weights ω^{inter} . Final weights: $\omega^* = W_{\text{intra}} \omega^{\text{inter}}$.

Empirical Proof: NCO vs. Classical Markowitz

Minimum Variance Portfolio (50 Assets)

- Markowitz Raw Allocation RMSE: 7.95×10^{-3}
- Ledoit-Wolf Shrinkage RMSE: 8.89×10^{-3}
- **NCO Allocation RMSE: 4.21×10^{-3}**
- **47.02% Allocation Error Reduction!**

Maximum Sharpe Ratio Portfolio

- Markowitz Raw Allocation RMSE: 7.02×10^{-2}
- Ledoit-Wolf Shrinkage RMSE: 6.54×10^{-2}
- **NCO Allocation RMSE: 3.17×10^{-2}**
- **54.83% Allocation Error Reduction!**

Summary: Key Takeaways on Optimal Clustering

Core Takeaways for Quantitative Asset Managers

- ➊ **Replace Heuristics with ONC:** Stop relying on subjective Elbow methods or naive k -means. ONC objectively identifies cluster count via Silhouette t -statistics (q).
- ➋ **Recursion Captures Sub-Structure:** Two-stage base + recursive subdivision uncovers nested hierarchical block structures in financial covariance matrices.
- ➌ **Solve Multicollinearity in Machine Learning:** Use ONC for Clustered Feature Importance (CFI) to neutralize substitution effects in Random Forests.
- ➍ **Stabilize Portfolio Weights:** Combine ONC with Nested Clustered Optimization (NCO) to achieve over 50% error reduction in Markowitz asset allocation.

Thank You!

Questions & Discussion

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